

Homework 2:

- 1. (25%) Group Discussion: P. 21 (5%), 32, 52
- 2. (70%) P. 22 shows an open-loop NN-power stage buck converter with specified circuit parameters.
 - (a) (30%) Please derive the equations of R_o of the equivalent circuit in p. 23 and the power efficiency as a function of circuit parameters. Then substitute parameter values into the derived equations to get R_o and efficiency values.
 - Use circuit simulation to verify the calculation results and reproduce the figures in p. 23, 24. Suggested simulation tool: LTspice or Virtuoso
 - (b) (40%) Please implement a driver circuit with functions of (1) Drive both switches properly. (2) Dead-time control to avoid shoot-through (3) Zero current detection for DCM operation.

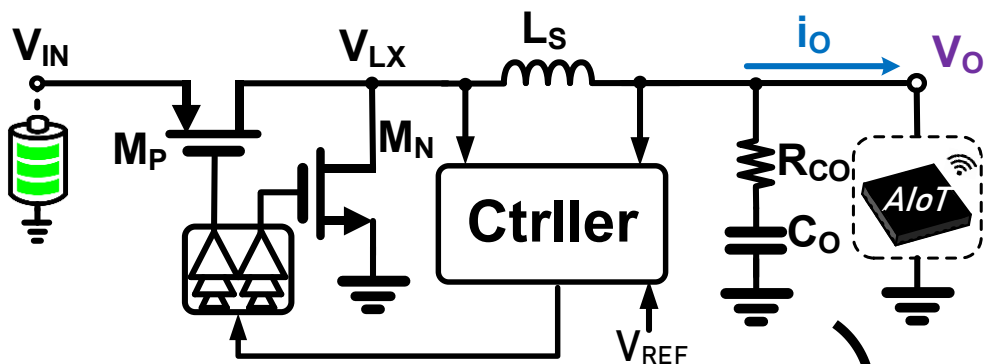


Homework 2:

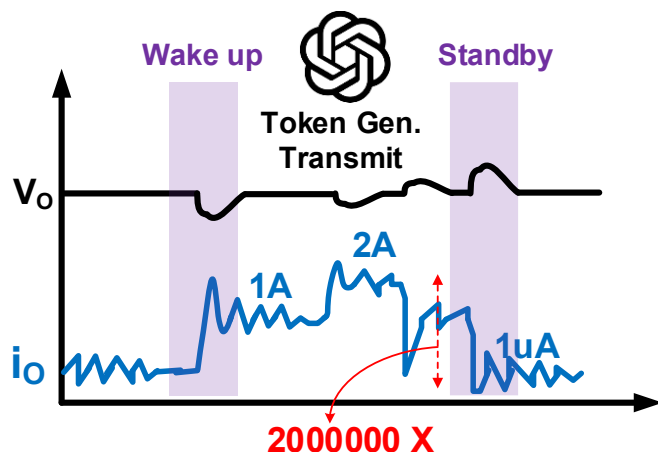
- The functions can be implemented by behavioral circuits and tested by providing a given duty signal.
 - (1) Show the implemented driver and switch circuit. Explain its operation. (2) Show the simulated waveforms including duty, gate drive signals v_{UG_DRIVE} and v_{LG_DRIVE} , VLX (VSW), i_L to prove the circuit achieves its function.
 - You got maximum 15 extra point if you can proof that your driver implementation have any advantage compared to the fixed dead-time, comparator ZCD detection, conventional level shifter, and bootstrap circuit in lecture notes.
3. (5%) PMIC for AIoT System (see next page). To achieve 85% efficiency at $I_o = 0.1$ mA, what is the maximum controller quiescent current budget? Hint: assume the converter loss is dominated by controller quiescent current.



3. PMIC for AIoT System



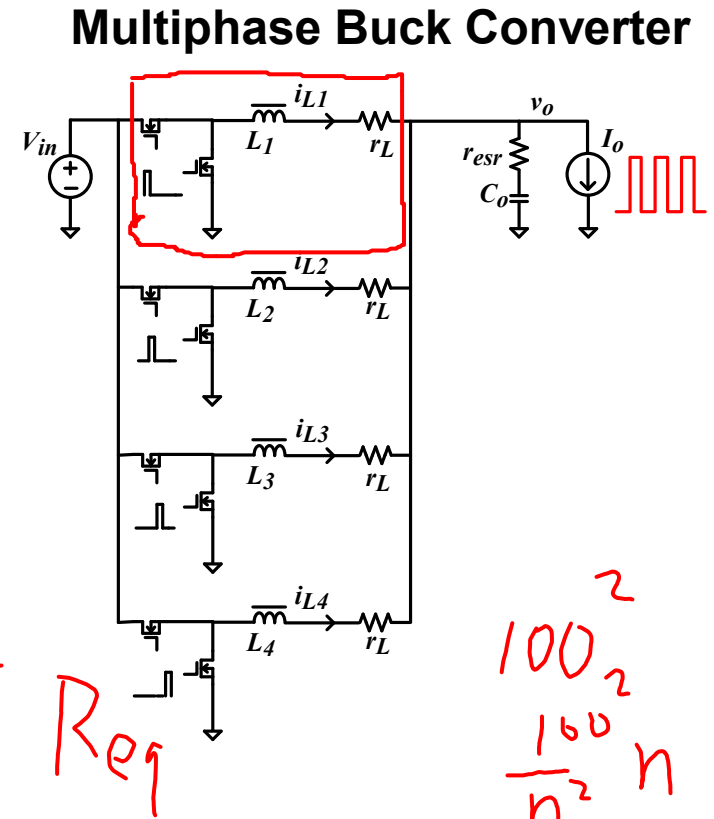
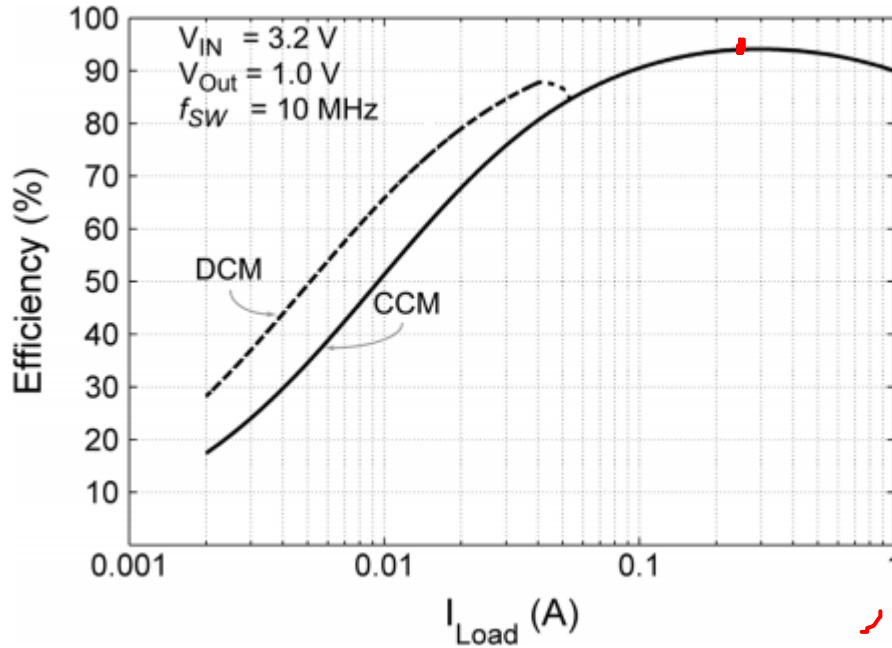
Cellphone, IoT devices



AI + IoT require PMIC with stringent specifications:

- Load transient for AIoT's wakeup, Token Gen. + Transmit ($\mu\text{A} \leftrightarrow \text{A}$)
- Ultra Wide Load Dynamic Range

P. 21: Why full-load efficiency also decrease?

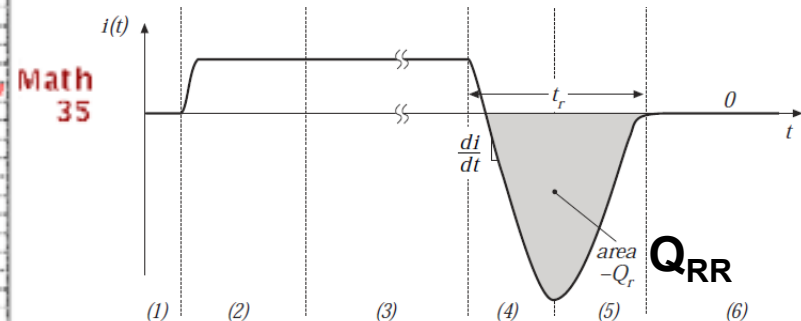
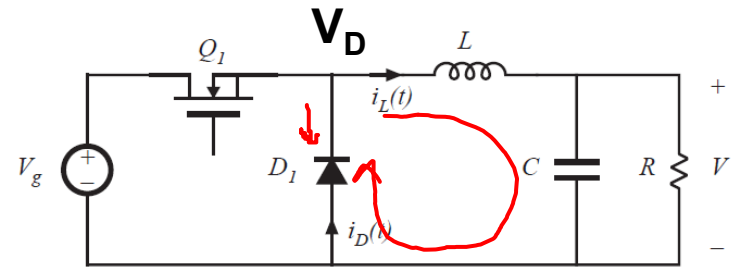
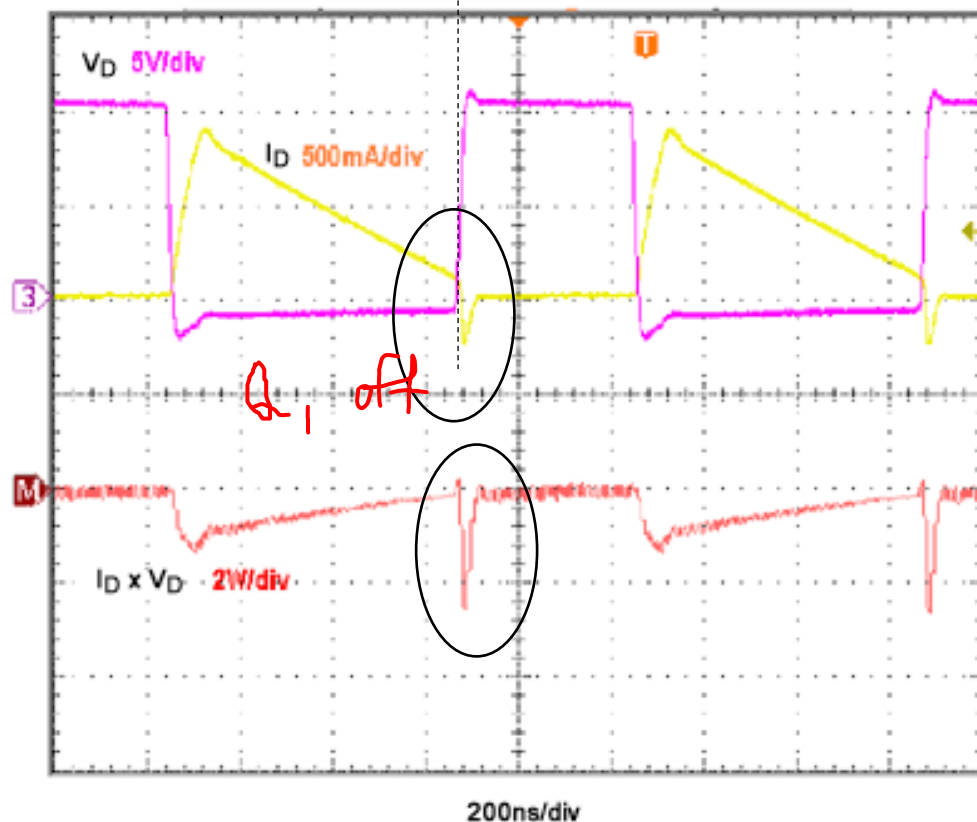


- Light-load efficiency decreases since switching loss is almost independent from load current
 - Why full-load efficiency also decrease?
- $$\eta = \frac{P_{out}}{P_{in}} = \frac{P_{in} - P_{loss}}{P_{in}}$$

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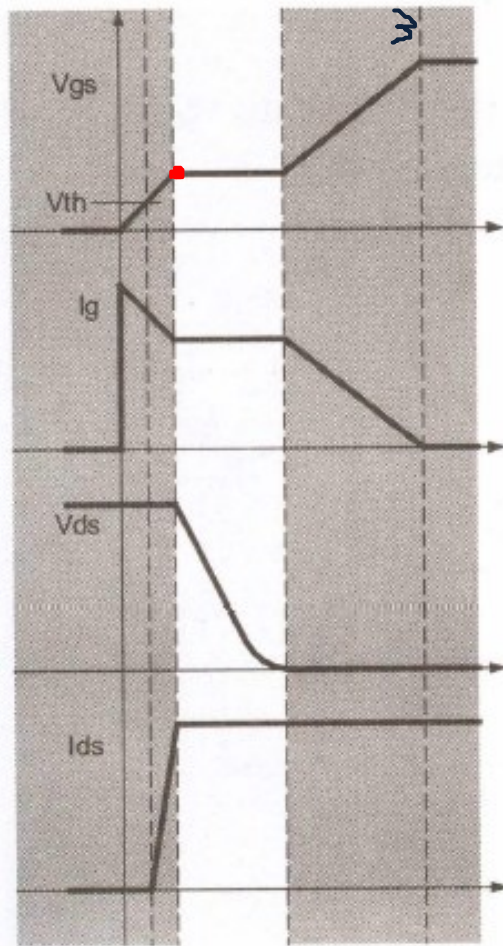
P. 32: Draw eq. ckt and diode bias status at each state, mark D.R.R. time and $iQ1$

Forward bias Reverse bias



- Reverse recovery happens when reverse-bias applied to a diode which has positive current flowing

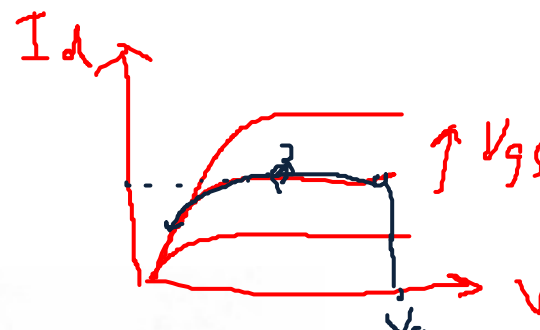
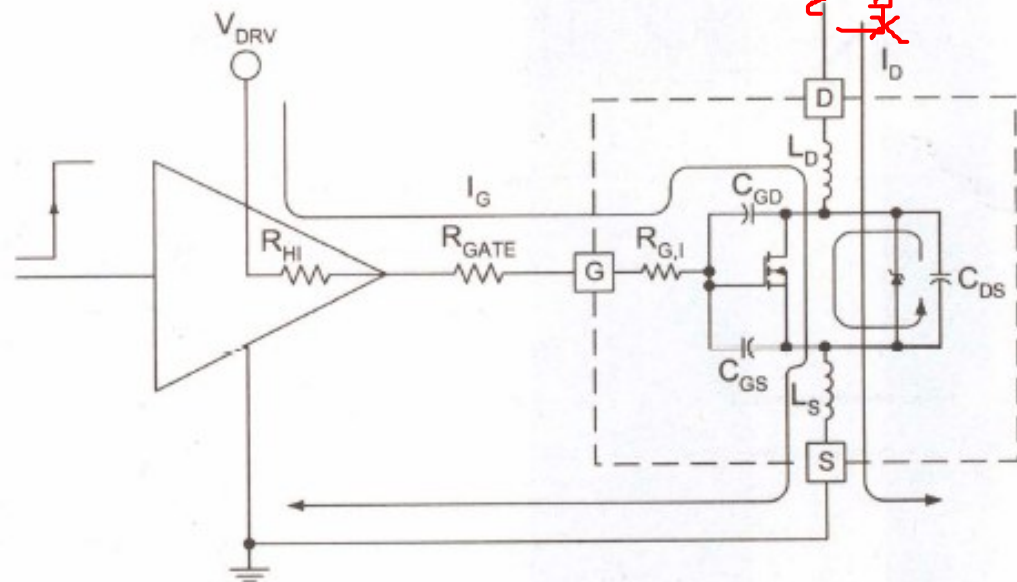
**P. 52: Explain waveform from Eq. ckt. Draw MOS operation curve at I_d - V_{ds} map
Calculate MOS VI overlap switching loss**



Power loss

Step 3.

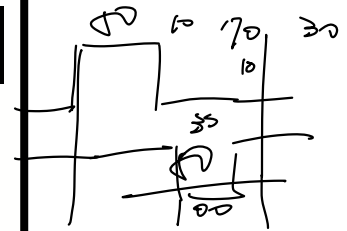
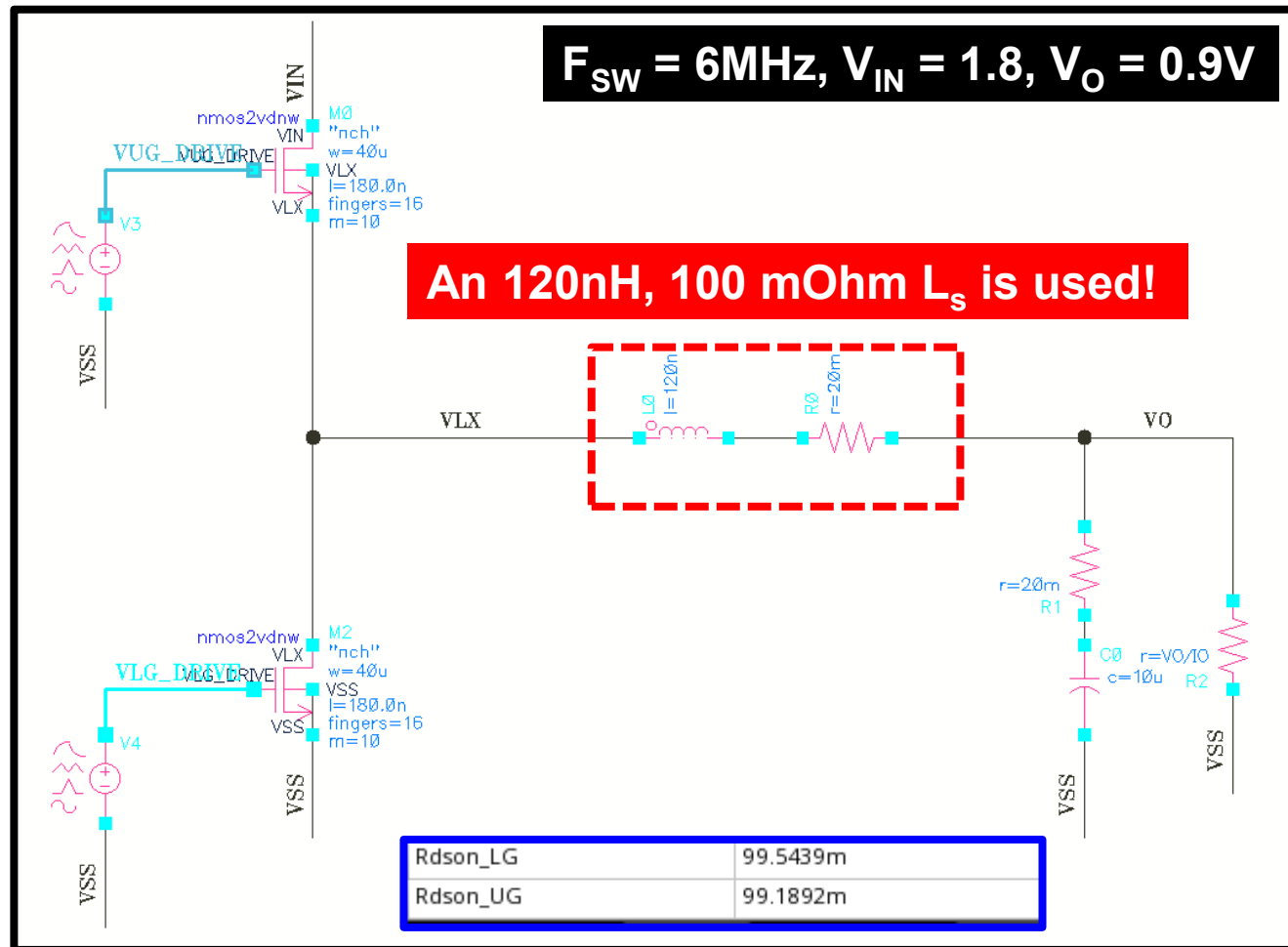
Discharge C_{GD} and C_{DS}
as V_{DS} falls close to GND.



$$i_c = C \frac{dv_c}{dt}$$

P. 22: Open-Loop NN-Power Stage Efficiency Simulation

Designing $R_{\text{dson}} = 100\text{m}\Omega$, $C_{\text{ISS}} = 14\text{pF}$, $C_{\text{OSS}} = 7\text{pF}$

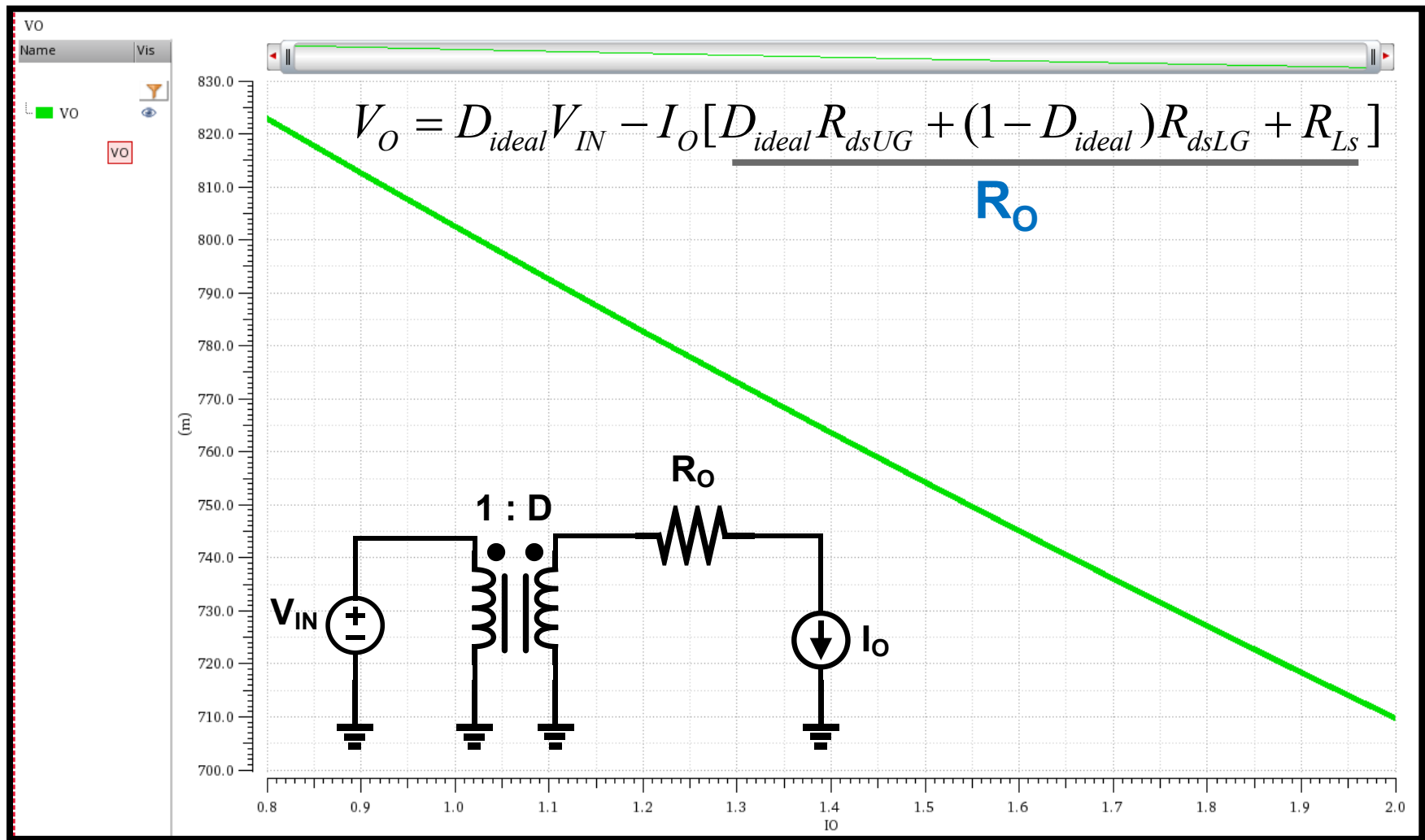


(of 60 fso)



83
167

P. 23: Open-Loop V_o v.s. I_o



P. 24: Efficiency Over Loading (0.8 – 2A)

